DATASCI 406 FA 26, Quiz 3.2

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Each question is worth 1 point. Circle one answer for each.

Question 1 (1 point)

In a sample of lead levels, several samples are tied at the median, and the jackknife SE of the median comes out to exactly 0. What does this tell you?

1. The median is estimated without error.
2. The leave-one-out medians take only a few distinct values.
3. The sample is too small. With more samples the jackknife SE would converge to the true SE of the median.
4. The factor $(n-1)/n$ should be left out for the median.

Question 2 (1 point)

The bootstrap-t interval uses the replicates

$$W_b^* = \frac{T_b^* - T}{\widehat{\mathrm{SE}}_b^*}, \quad b = 1, \ldots, B.$$

Which SE goes in the denominator?

1. The jackknife SE from the original sample, the same for every b.
2. The standard deviation of all B bootstrap replicates $T_1^*, \ldots, T_B^*$.
3. The bootstrap SE from the original sample, the same for every b.
4. The jackknife SE computed from bootstrap sample b itself.

Question 3 (1 point)

Polynomials of degree $d = 1, \ldots, 10$ are fit to the same data. The training MSE falls with every increase in d. Why does this not show that $d = 10$ predicts best?

1. Each fit is scored on the same points that were used to fit it.
2. The training MSE has Monte Carlo error, so small differences are noise.
3. A model with more parameters always predicts new points worse.
4. Leave-one-out MSE always chooses $d = 1$.

Question 4 (1 point)

A bike share predicts tomorrow's rentals y with $\hat{y}$ to decide how many bikes to put out. Running out of bikes is a much worse mistake than leaving some unused. With $e = y - \hat{y}$, three losses are

$$\text{squared: } e^2, \qquad \text{absolute: } |e|, \qquad \text{pinball: } L_\tau(e) = \begin{cases} \tau\, e & e > 0, \\ (1-\tau)(-e) & e \le 0. \end{cases}$$

Which loss should cross-validation use?

1. Squared error.
2. Absolute error.
3. Pinball loss with $\tau = 0.9$.
4. Pinball loss with $\tau = 0.1$.